

Lab 4 – KNN

DS3010: Introduction to Machine Learning Lab

Timing: 02:00 PM to 04:45 PM

Max Marks:10

Instructions

1. Submit one .ipynb file containing all answers. The name should be [student_name]_[rollno]_[lab].ipynb
2. Write the questions in separate text blocks before the answers.
3. Outputs for all sub-questions should be given and the code should be executable.
4. Write justifications for your choices where needed.
5. Ensure that all plots include clear labels and legends for better interpretation.
6. **Use of generative AI tools (such as ChatGPT, Gemini, etc.) is strictly prohibited. Any submission found to contain AI-generated or plagiarized content will receive a score of zero, and disciplinary action.**

The following code can be used to load mnist dataset

You can use plt.imshow() to display images

```
from sklearn.datasets import fetch_openml

mnist = fetch_openml('mnist_784', version=1, as_frame=False)

X = mnist.data
y = mnist.target.astype(int)
```

```
plt.imshow(X_train[0].reshape(28, 28), cmap="gray")
```

KNN

1. Dataset Loading and Preprocessing

Load the MNIST dataset and split it into training, and test sets(70-30). Inspect the shape of the dataset and display a few sample images with their corresponding labels. Normalize the pixel values and investigate whether feature scaling affects KNN performance.

You may find the following link useful

<https://medium.com/@adnan.mazraeh1993/feature-scaling-from-basic-to-super-advanced-1-3519644eb5c8>
Feature Scaling: From Basic to Super Advanced

- What is the dimensionality of each input sample?
- Compare the performance of KNN with and without feature scaling.
- Does feature scaling significantly matter for MNIST? Explain why.

2. Implement KNN from Scratch

Implement the K-Nearest Neighbors (KNN) algorithm from scratch without using a pre-built KNN classifier. Your implementation should:

- (a) Calculate the Euclidean distance between samples.
- (b) Find the k nearest training samples.
- (c) Predict the class using majority voting.

Evaluate your implementation on a subset of MNIST (**try taking 5000 train points and 200 test points**) and report the classification accuracy and confusion matrix.

3. Choosing the Value of k

Evaluate KNN using multiple values of k , for example

$$k \in \{1, 3, 5, 7, 9, 11\}.$$

Plot the test error against k for

$$k = 1, 2, \dots, N.$$

Identify the value of k that gives the lowest test error and briefly explain why very small and very large values of k can lead to overfitting and underfitting, respectively.

4. Investigate Distance Metrics

Compare KNN using different distance metrics:

- Euclidean distance
- Manhattan distance

You may find the following link useful

<https://www.geeksforgeeks.org/data-science/manhattan-distance/> Manhattan Distance

Evaluate both methods using the same value of k and compare their test accuracy. Briefly explain how the choice of distance metric affects KNN predictions and which metric performs better on the MNIST dataset.

5. Final Analysis

Summarize the results of all experiments in a table containing the feature scaling method, number of features, best value of k , test accuracy, and prediction time.

Finally, answer the following:

- What value of k performed best?
- Did feature scaling improve KNN performance on MNIST?
- What are the main advantages and disadvantages of KNN for image classification?